

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

I V Sunil Kumar/192425292 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer:

The large difference between the **training MSE (0.85)** and **testing MSE (5.20)** indicates that the neural network performs very well on the training data but poorly on unseen data. This is a strong sign of **overfitting**, where the model memorizes the training data instead of learning general patterns.

Possible Reasons

1. **Overfitting**
 - The network has learned the training examples too closely, including noise.
 - Results in low training error but high testing error.
2. **Complex Model**
 - Too many hidden layers or neurons compared to the size of the dataset.
 - High model complexity reduces generalization ability.
3. **Insufficient Training Data**
 - A small or non-representative training dataset prevents the model from learning diverse patterns.
4. **Poor Data Quality**
 - Noisy, inconsistent, or irrelevant features in the training data can reduce test performance.
5. **Improper Data Split**
 - Training and testing datasets may have different distributions, causing poor generalization.
6. **Lack of Regularization**
 - Without techniques such as dropout or L2 regularization, the model is more likely to overfit.

Corrective Measures

- **Reduce Model Complexity**
 - Use fewer hidden layers or neurons.
 - Remove unnecessary parameters.
- **Apply Regularization**
 - Use L1/L2 regularization or dropout to prevent overfitting.
- **Increase Training Data**
 - Collect more data or apply data augmentation where appropriate.
- **Use Early Stopping**
 - Stop training when validation error starts increasing instead of continuing until training error is minimized.
- **Perform Cross-Validation**
 - Use techniques like k-fold cross-validation to obtain a more reliable estimate of model performance.
- **Improve Data Preprocessing**
 - Normalize features, remove noisy or irrelevant data, and handle missing values.

Conclusion

The low **training MSE (0.85)** and much higher **testing MSE (5.20)** indicate that the neural network is **overfitting** and has poor generalization. Reducing model complexity, applying regularization, increasing data quality and quantity, using early stopping, and validating the model with cross-validation can significantly improve its performance on unseen data.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided